

Index Theory

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ABSTRACT. I follow Gilkey's book

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Pseudo Differential Operators

1. Fourier Transforms, Schwartz class, Sobolev Spaces

1.1. Notation. We work in $\mathbb{R}^m$. For $\alpha_i \geq 0$

$$\begin{aligned}\alpha &= (\alpha_1, \dots, \alpha_m), \quad |\alpha| = \alpha_1 + \dots + \alpha_m, \\ \alpha! &= \alpha_1! \cdots \alpha_m!, \quad x^\alpha = x_1^{\alpha_1} \cdots x_m^{\alpha_m} \\ \partial^\alpha &= \partial_x^\alpha = \left(\frac{\partial}{\partial x_1} \right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_m} \right)^{\alpha_m}, \quad D^\alpha = D_x^\alpha = (-i)^{|\alpha|} \partial_x^\alpha.\end{aligned}$$

Taylor's formula: For $f \in C^\infty$,

$$f(x) = \sum_{|\alpha| \leq n} \frac{1}{\alpha!} \partial^\alpha f(x_0) (x - x_0)^\alpha + O(|x - x_0|).$$

Schwartz class:

$$\mathcal{S} = \{f : \mathbb{R}^m \rightarrow \mathbb{C} \mid f \in C^\infty, \forall n \forall \beta \ |D^\beta f(x)| \leq C_{n,\beta} (1 + |x|)^{-n}\},$$

where $C_{n,\beta}$ depend on f .

To simplify the FT formulae, Lebesgue measure on $\mathbb{R}^m$ is normalised by $(2\pi)^{-m/2}$, so that

$$\int e^{-\frac{1}{2}|x|^2} dx = 1.$$

Convolution: Define

$$(f * g)(x) := \int f(x - y) g(y) dy = \int f(y) g(x - y) dy.$$

This defines an *associative and commutative multiplication*.

PROPOSITION 1.1. *If $f \in \mathcal{S}$, $\int f = 1$, $f_\epsilon(x) := \epsilon^{-n} f(\frac{x}{\epsilon})$, and $g \in \mathcal{S}$, then*

$$f_\epsilon * g \rightarrow g \quad \text{uniformly as } \epsilon \rightarrow 0.$$

One can use this approach to show any $g \in L^p$ can be approximated in the L^p norm by a C^∞ function of compact support.

1.2. Definition of FT on $\mathcal{S}$ and basic properties. Define

$$\hat{f}(\xi) = \int e^{-ix \cdot \xi} f(x) dx.$$

It follows, using integration by parts and dominated convergence for the first line, that ^{1 2}

$$\begin{aligned}D_x^\alpha f(x) &\leftrightarrow \xi^\alpha \hat{f}(\xi), \quad x^\alpha f(x) \leftrightarrow D_\xi^\alpha \hat{f}(\xi), \\ (1) \quad f(x - x_0) &\leftrightarrow e^{-ix_0 \cdot \xi} \hat{f}(\xi), \quad e^{ix \cdot \xi_0} f(x) \leftrightarrow \hat{f}(\xi - \xi_0).\end{aligned}$$

In particular, it follows $\hat{f} \in \mathcal{S}$.

Calculation shows

$$e^{-\frac{1}{2}|x|^2} \leftrightarrow e^{-\frac{1}{2}|\xi|^2}.$$

PROPOSITION 1.2. $f \leftrightarrow \hat{f}$ is a bijection on $\mathcal{S}$. In fact

$$f(x) = \int e^{ix \cdot \xi} \hat{f}(\xi) d\xi = \hat{\hat{f}}(-x) =: Tf(x).$$

¹Typo in book

²Where “ $\leftrightarrow$ ” means the right side is the FT of the left side.

PROOF. Suppose $f \in \mathcal{S}$.

First suppose $f(0) = 0$. Use

$$f(x) = \int_0^1 f(tx) dt \quad \text{to get} \quad f(x) = \sum_j x_j h_j(x) \quad \text{where } h_j \in \mathcal{S},$$

and so³

$$\hat{f} = -i \sum_j \frac{\partial \hat{h}_j}{\partial \xi_j}, \quad \therefore Tf(0) = \int \hat{f} = 0 = f(0).$$

For general f subtract a multiple of $e^{-\frac{1}{2}|x|^2}$ and note that T is the identity on $e^{-\frac{1}{2}|x|^2}$. It follows $Tf(0) = f(0)$ for any $f \in \mathcal{S}$.

To get equality at any $x \in \mathbb{R}^m$ use (1). $\square$

We also have

$$\begin{aligned} f * g &\leftrightarrow \hat{f} \cdot \hat{g}, & f \cdot g &\leftrightarrow \hat{f} * \hat{g} \quad \forall f, g \in \mathcal{S}, \\ (f, g) &= (\hat{f}, \hat{g}) \quad \forall f, g \in L^2(\mathbb{R}^m). \end{aligned}$$

The last gives a unitary map on $L^2(\mathbb{R}^m)$ and is known as *Plancherel's theorem*.

1.3. Sobolev Spaces.

Definition: For $f \in \mathcal{S}$ and $s \in \mathbb{R}$ let

$$|f|_s^2 = \int (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 d\xi.$$

Note $(1 + |\xi|^2)^s$ and $(1 + |\xi|)^{2s}$ give equivalent norms. $\mathcal{H}_s(\mathbb{R}^m)$ is the completion of $\mathcal{S}$ in this norm.

Equivalent spaces: $\mathcal{H}_0 \approx L^2$ by Plancherel's theorem. $\mathcal{H}_s \approx L^2$ if the measure $(1 + |\xi|^2)^{s/2}$ is used.

Connection with derivatives: s counts the number of derivatives in the sense

$$\sum_{|\alpha| \leq n} \int |D^\alpha f(x)|^2 = \sum_{|\alpha| \leq n} \int |\xi^\alpha \hat{f}(\xi)|^2 \sim \|f\|_s^2,$$

since

$$c_1(1 + |\xi|^2)^n \leq \sum_{|\alpha| \leq n} |\xi^\alpha|^2 \leq c_2(1 + |\xi|^2)^n.$$

Continuous map: The map $D^\alpha \mathcal{H}_s \rightarrow \mathcal{H}_{s-|\alpha|}$ is continuous.

Higher derivatives imply continuity: If $s > k + m/2$ and $\mathcal{H}_s \subset C^k$ and

$$|f|_{\infty, k} \leq C|f|_s.$$

Dense subset: C_0^∞ is dense in $\mathcal{H}_s$.

Rellich's lemma: Suppose $(f_n)_{n \geq 1} \in \mathcal{S}$ with uniform compact support, and uniform $\mathcal{H}_s$ bound. Then if $t < s$, some subsequence converges in $\mathcal{H}_t$.

Natural dual pairing: The map $(f, g) \mapsto \int f \bar{g}$ extends to a map $\mathcal{H}_s \times \mathcal{H}_{-s} \rightarrow \mathbb{C}$ which identifies $\mathcal{H}_{-s}$ with $\mathcal{H}_s^*$. In fact

$$\begin{aligned} |(f, g)| &\leq |f|_s |g|_{-s}, \\ |f|_s &= \sup\{|(f, g)| : |g|_{-s} = 1\}. \end{aligned}$$

For the proof of (1) note

$$|(f, g)| = |(\hat{f}, \hat{g})| = \int \hat{f}(\xi) (1 + |\xi|^2)^{s/2} \bar{\hat{g}}(\xi) (1 + |\xi|^2)^{-s/2},$$

and use Cauchy Schwartz. For (2) choose g so $\hat{g} = \hat{f}(1 + |\xi|^2)^s$ and normalise.

Peter Paul principle: If $s > t > u$ and $\epsilon > 0$ then

$$(1 + |\xi|)^{2t} \leq \epsilon(1 + |\xi|)^{2s} + C(\epsilon)(1 + |\xi|)^{2u},$$

³missing “—” in book.

and so

$$|f|_t \leq \epsilon |f|_s + C(\epsilon) |f|_u.$$

Vector-valued case: All the preceding generalises readily to functions with values in a k dimensional space.